

PH102

After Mid Semester Exams

by
Prof. Subhash Thota & Dr. Pankaj Kumar Mishra

Department of Physics, IIT Guwahati

Chapter-7

Electromagnetic Induction

Continuation of earlier class on the topic
“ Drift Velocity”

&

Comments on the Microscopic Aspects
of Electrical Conduction in Metals, (i.e.
good conductors)

Thermal Velocity of Electrons

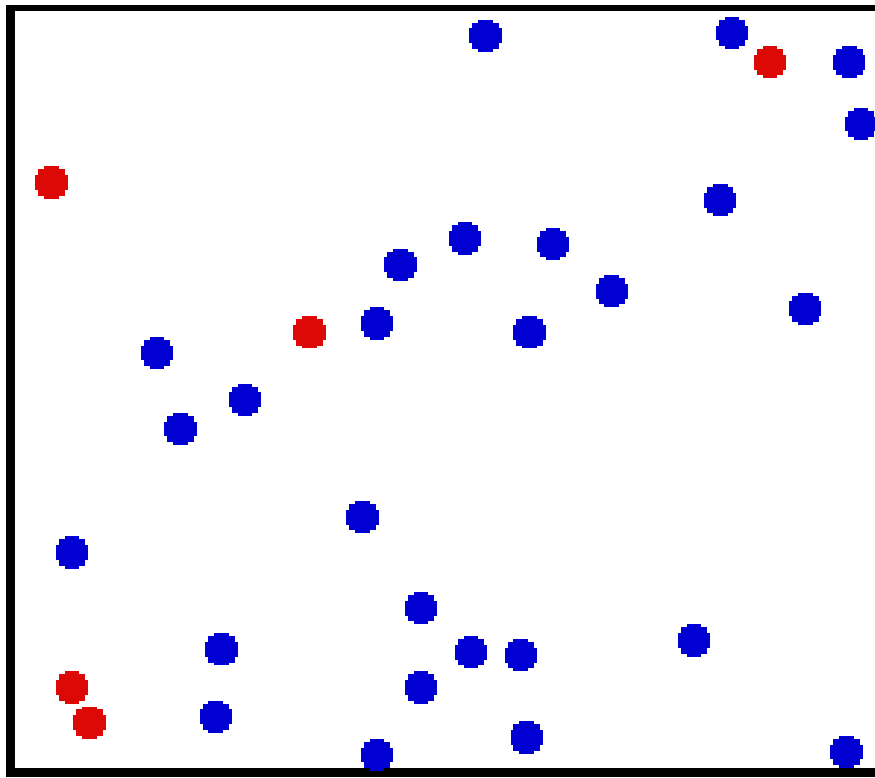
In any metal, the “free” electrons exist as a “gas” of electrically charged particles inside the bulk material of the conductor. If the entire block of metal is in thermal equilibrium with its surrounding local environment, then the “free” electron “gas” is also in thermal equilibrium.

The equipartition (of energy) theorem says that for each degree of freedom associated with such particles, there is associated $\frac{1}{2} \mathbf{k_B T}$ of thermal energy associated with it (in equilibrium), where Boltzmann’s constant, $\mathbf{k_B} = 1.381 \times 10^{-23}$ Joules/Kelvin

A “free” electron in a metal has 3 kinetic degrees of freedom

$$\langle v_x \rangle, \langle v_y \rangle, \langle v_z \rangle$$

The $\hat{x}$, $\hat{y}$, and $\hat{z}$ components of the free electron’s mean (or average) thermal velocity



Thermal equilibrium. The condition under which two substances in physical contact with each other exchange no heat energy. Two substances in **thermal equilibrium** are said to be at the same temperature.

The kinetic energy of a particular molecule can fluctuate wildly, but the equipartition theorem allows its *average* energy to be calculated at any temperature. Equipartition also provides a derivation of the ideal gas law, an equation that relates the pressure, volume and temperature of the gas. (In this diagram five of the molecules have been colored red to track their motion; this coloration has no other significance.)

....the frequent collisions electrons make as they pass down the wire...

Then the mean (or average) kinetic energy of a free electron is

$$K.E. = \frac{1}{2}m\langle v_x \rangle^2 + \frac{1}{2}m\langle v_y \rangle^2 + \frac{1}{2}m\langle v_z \rangle^2$$

By the Equipartition theorem (Page 289):

$$K.E. = E_{thermal} = \langle E_x^{thermal} \rangle + \langle E_y^{thermal} \rangle + \langle E_z^{thermal} \rangle$$

$$K.E. = E_{thermal} = \frac{1}{2}k_B T + \frac{1}{2}k_B T + \frac{1}{2}k_B T$$

The **Boltzmann constant**
($k_B = 1.38 \times 10^{-23} \text{ J/K}$)

Ludwig E Boltzmann



$$K.E. = \frac{1}{2}m\langle v_e \rangle^2 = \frac{3}{2}k_B T$$

20 February 1844 – 5 September 1906

The average thermal speeds of “free” electrons in metals are therefore

$$v = \sqrt{\frac{3k_B T}{m}}$$

Thus, for $T = 300\text{K}$ (at room temperature 27°C) and $m_e = 9.1 \times 10^{-31} \text{ kg}$, typical thermal electron speeds at room temperature are:

$$v(\text{at room temperature}) = \sqrt{\frac{3 \times 1.381 \times 10^{-23} \times 300}{9.1 \times 10^{-31}}}$$

$$v(\text{at room temperature}) = 1.2 \times 10^5 \text{ meters/sec}$$

$$v(\text{at room temperature}) = 1.2 \times 10^5 \text{ meters/sec}$$

However, This is not the speed at which electrons conduct electrical power e.g. when flowing through a wire.

The “free” conduction electrons scatters with atoms (and each other) in the metal, with a mean free path, $\lambda_{mfp} \equiv$ average distance between successive collisions/scatterings. The average time between successive collisions / scatterings is

$$\langle \tau \rangle = \frac{\lambda_{mfp}}{\langle v_{thermal} \rangle}$$

Drift Velocity

Acceleration of free electrons inside the wire

Free electrons inside a conducting wire are accelerated by longitudinal Electric field E :

$$\vec{F} = q\vec{E}$$

$$\vec{F} = -e\vec{E}$$

$$m\vec{a}_e = -e\vec{E}$$

$$\vec{a}_e = -\frac{e\vec{E}}{m}$$

However, electrons *elastically* scatter off of the atoms inside the conductor – loose kinetic energy to the atoms (which ultimately winds up as heat) – electrons *drift* through conductor with an average terminal velocity of:

$$\langle \vec{v}_d \rangle = \frac{1}{2} a \langle \tau \rangle$$

$\langle \tau \rangle = \text{mean/average time between successive collisions}$

$$\langle \tau \rangle = \frac{\lambda_{mfp}}{\langle v_{thermal} \rangle}$$

$\lambda_{mfp} = \text{mean free path}$

$\langle v_{thermal} \rangle = \text{average thermal speed of electrons}$

Thus

$$\langle \vec{v}_d \rangle = \frac{1}{2} a \langle \tau \rangle = \frac{1}{2} a \frac{\lambda_{mfp}}{\langle v_{thermal} \rangle}$$

If there are n atoms/molecules per unit volume and a fraction f of free electrons per atom/molecule

Then

$$J_{free} = (nf)q\langle \vec{v}_d \rangle$$

Refer
Previous Lecture

$$J_{free} = (n_q)q\langle \vec{v}_d \rangle$$



$$\begin{aligned} I &= nqv_d A \\ J &= \frac{I}{A} = nqv_d \\ \vec{J} &= nq\vec{v}_d \end{aligned}$$

Where n_q is the number density of free electrons = $\mathbf{n} \times \mathbf{f}$

$$J_{free} = (n_q)q\langle\vec{v}_d\rangle$$



$$J_{free} = (n_q)q\frac{1}{2}a\frac{\lambda_{mfp}}{\langle v_{thermal}\rangle}$$



$$\langle\vec{v}_d\rangle = \frac{1}{2}a\frac{\lambda_{mfp}}{\langle v_{thermal}\rangle}$$

$$J_{free} = (n_q)q\frac{1}{2}\frac{q\vec{E}}{m}\frac{\lambda_{mfp}}{\langle v_{thermal}\rangle}$$



$$\vec{a}_e = -\frac{e\vec{E}}{m}$$

$$J_{free} = \frac{1}{2}\frac{n_q q^2 \lambda_{mfp}}{\langle v_{thermal}\rangle} \frac{\vec{E}}{m}$$

Let us calculate the free electron number density n_e for e.g. Copper (Cu):

Avogadro's Number N_A is given by

$$N_A = 6.022 \times 10^{23} \quad \text{atoms per gm-mol}$$

Copper has an atomic weight of $A_{\text{Cu}} = 63.54$

It means 63.54 grams of copper contains 6.022×10^{23} atoms of copper

We need the number density of free electrons in copper ??

$$\therefore 1 \text{ gm of copper contain} = \frac{N_A}{A_{Cu}} = \frac{6.022 \times 10^{23}}{63.54} = 9.477 \times 10^{21}$$

Nominally, atomic copper has an electron valency of one, thus in a copper metal, for every copper atom, there is associated with it one “free” electron. Thus, in one gram of copper, there are 9.477×10^{21} free electrons.

The mass density of copper

$$\rho_{Cu} = 8.95 \text{ grams/cm}^3$$

Then the number density of free electrons in copper is given by

$$n_e^{Cu} = 9.477 \times 10^{21} \times \rho_{Cu}$$

$$n_e^{Cu} = 9.477 \times 10^{21} (\text{electrons/g}) \times 8.95 (\text{g/cm}^3)$$

$$n_e^{\text{Cu}} = 8.482 \times 10^{22} (\text{electrons/cm}^3)$$

$$n_e^{\text{Cu}} = 8.482 \times 10^{28} (\text{electrons/m}^3)$$

Suppose we have a copper wire carrying a steady free current of $I_{\text{free}} = 1$ Ampere, the copper wire e.g. has a cross-sectional area of $A_{\perp} = 1 \text{ mm}^2 = 10^{-6} \text{ m}^2$.

Then

$$\langle \overrightarrow{v_d} \rangle = \frac{\overleftarrow{J_{\text{free}}}}{n_e q}$$

$$\langle \overrightarrow{v_d} \rangle = \frac{\overleftarrow{I_{\text{free}}/A_{\perp}}}{n_e q}$$

$$q = 1.602 \times 10^{-19} \text{ C}$$

$$\langle \overrightarrow{v_d} \rangle = \frac{\frac{1.0}{10^6}}{8.482 \times 10^{28} \times 1.602 \times 10^{-19}}$$

$$\langle \overrightarrow{v_d} \rangle = 7.359 \times 10^{-5} \frac{m}{sec}$$

$$\langle \overrightarrow{v_d} \rangle = 73.6 \frac{\mu m}{sec}$$

In one hour

$$\Delta t = 60 \times 60 = 3600 \text{ sec}$$

The free electrons drifting in a copper conductor will have traveled a distance of:

$$\Delta L = \langle v_d \rangle \Delta t = 0.265 \text{ m} = 26.5 \text{ cm}$$

This spectacularly slow drift velocity $\langle \vec{v}_d \rangle \sim 75 \mu\text{m}/\text{sec}$ doesn't square at all with our everyday experience, e.g. of turning on a light which is incredibly fast – essentially instantaneous, on human-perceived time-scales (e.g. a few milliseconds) (fast LEDs – light emitting diodes – can turn on/off in a few Nano-seconds (10^{-9} sec)).

Note that $\langle \vec{v}_d \rangle \sim 75 \mu\text{m}/\text{sec} \ll c = 3 \times 10^8 \text{ m/s}$ (**speed of light in a vacuum**)

Typical electrical signals in conducting wires can easily propagate at $\sim 50\%$ speed of light c

$$v_{prop} \cong 1.5 \times 10^8 \text{ m/s}$$

Thus $\langle \vec{v}_d \rangle$ does not correspond in any way to v_{prop} !!!

7.2. Electromagnetic Induction in detailed

Experiment 1

A loop is pulled of the wire to the right through a magnetic field (Fig 1). A current flowed in the loop.

Experiment 2

He moved the magnet to the left, holding the loop still (Fig 2). Again, a current flowed in the loop.

Experiment 3

With both the loop and the magnet at rest (Fig 3), he changed the strength of the field (he used an electromagnet, and varied the current in the coil). Once again, current flowed in the loop.

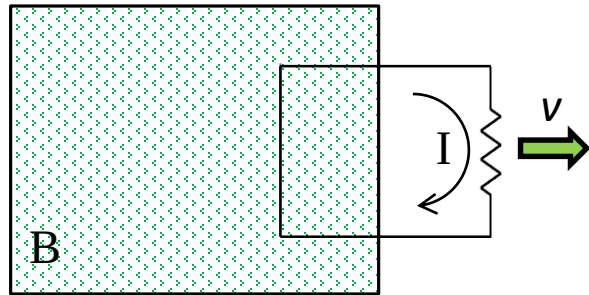


Figure 1

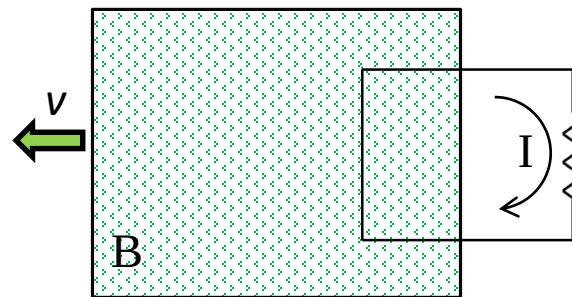


Figure 2

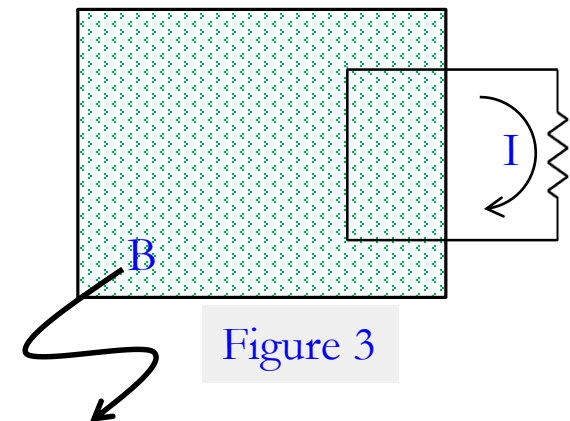


Figure 3

$$\varepsilon = - \frac{d\Phi}{dt}$$

Changing magnetic field

The first experiment, of course, is an example of motional electro-motive-force (emf), conveniently expressed by the flux rule:

$$\varepsilon = -\frac{d\Phi}{dt}$$

I don't think it will surprise you to learn that exactly the same emf arises in Experiment 2, all that really matters is **the relative motion of the magnet and the loop**.

For if the loop moves, it's a magnetic force that sets up the emf, but if the loop is stationary. The force cannot be magnetic — **Stationary charges experience** no magnetic forces. In that case, what is **responsible** ? What sort of field exerts a force on charges at rest? Well, electric fields do, of course, but in this case there doesn't seem to be any electric field in sight.

Faraday had an ingenious inspiration:

A changing magnetic field induces an electric field

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

It is this "induced" electric field that accounts for the emf in **Experiment 2**.

Indeed, if (as Faraday found empirically) the emf is again equal to the rate of change of the flux

$$\varepsilon \equiv \oint \vec{E} \cdot d\vec{l}$$

$$\varepsilon = -\frac{d\Phi}{dt}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt} \longleftarrow \Phi = \int \vec{B} \cdot d\vec{a}$$

Then E is related to the change in B by the equation

$$\Phi = \int \vec{B} \cdot d\vec{a}$$

This is Faraday's
law, in integral
form

$$\oint \vec{E} \cdot d\vec{l} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a}$$

Applying Stokes' theorem

This is Faraday's law, in integral form. We can convert it to differential form by applying Stokes' theorem:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

In the static case

$$\vec{B} = \text{Constant}$$

$$\frac{\partial \vec{B}}{\partial t} = 0$$

$\therefore$

$$\nabla \times \vec{E} = 0$$

Hence the Faraday's law reduces to the old rule $\nabla \times \vec{E} = 0$

Or the integral form

$$\oint \vec{E} \cdot d\vec{l} = 0$$

In Experiment 3 the magnetic field changes for entirely different reasons, but according to Faraday's law an electric field will again be induced, giving rise to an emf $-\frac{d\Phi}{dt}$. Indeed, one can subsume all three cases (and for that matter any combination of them) into a kind of **universal flux rule**:

Whenever (and for whatever reason) the magnetic flux through a loop changes, an emf

$$\varepsilon = -\frac{d\Phi}{dt}$$

Will appear to the loop

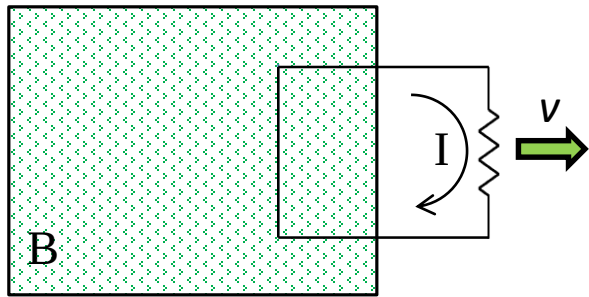


Figure 1

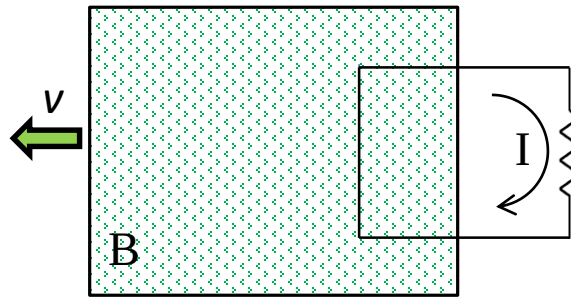


Figure 2

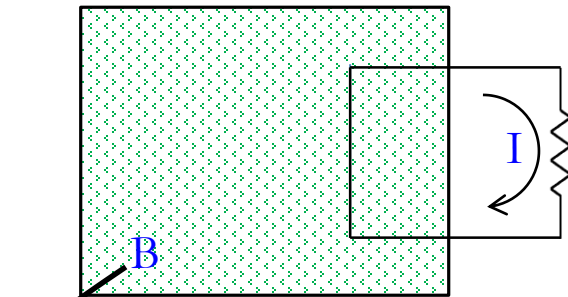


Figure 3

Changing magnetic field

Example 1: A long cylindrical magnet of length ' L ' and radius ' a ' carries a uniform magnetization ' M ' parallel to its axis. It passes at constant velocity ' v ' through a circular ring of slightly larger diameter (Figure 1). Graph the emf induced in the ring, as a function of time.

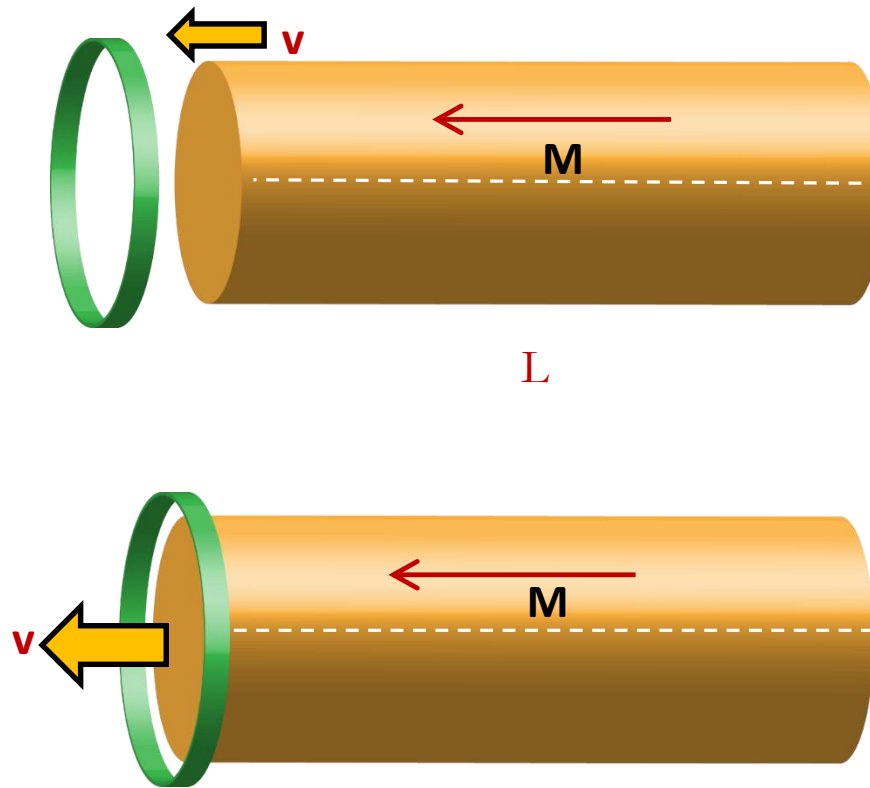


Figure 1

Solution: The magnetic field is the same as that of a long solenoid with surface current:

$$K = nI = I_{\text{enclosed}}$$

$$\vec{K}_b = M\hat{\phi}$$

See Griffiths page no. 227 , [Example 5.9](#)

For Solenoid

$$\vec{B} = \mu_0 n I \hat{z}$$

Inside

$$\vec{B} = 0$$

Outside

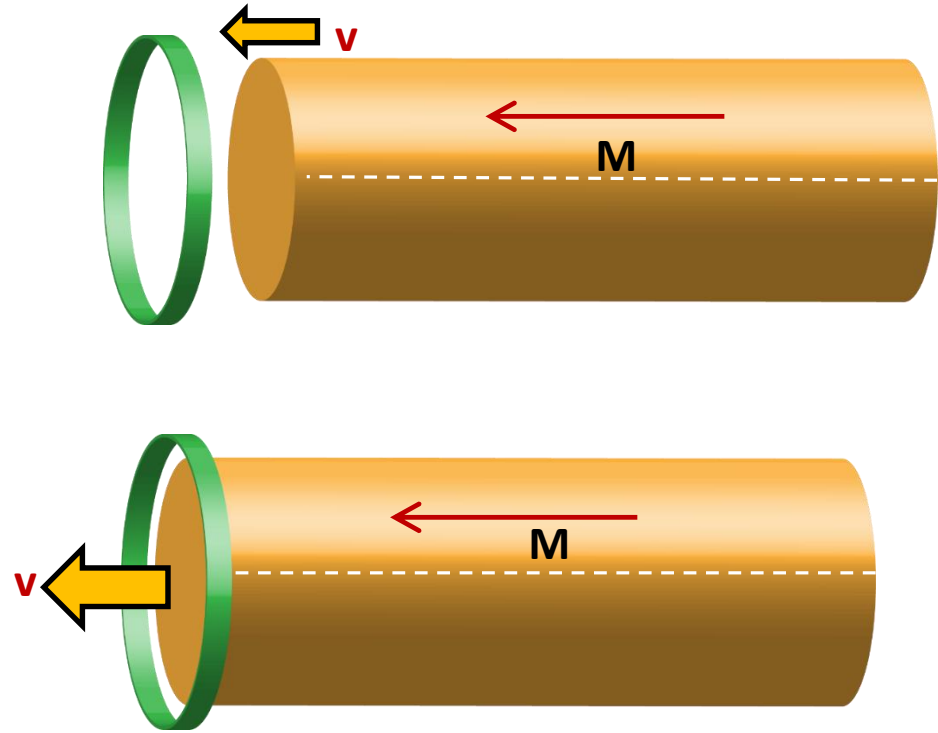


Figure 1

So for the present condition the field inside is

$$\vec{B} = \mu_0 \vec{M} \quad \boxed{\text{Except near the ends}}$$

The flux through the ring is zero when the magnet is far away.

As the leading end passes through it builds up to a maximum flux of

$$\Phi_{\max} = \mu_0 M \pi a^2$$

And it drops back to zero as the trailing end emerges.

Since the emf is minus the derivative of ' Φ ' with respect to time

$$\varepsilon = -\frac{d\Phi}{dt}$$

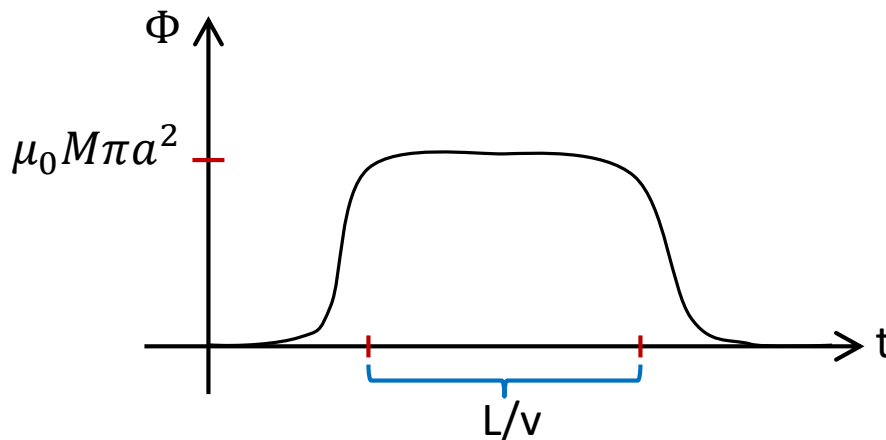


Figure 2a

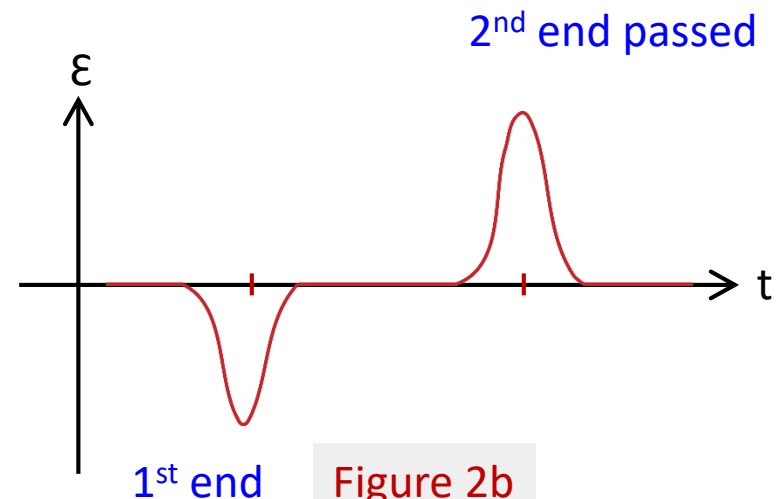
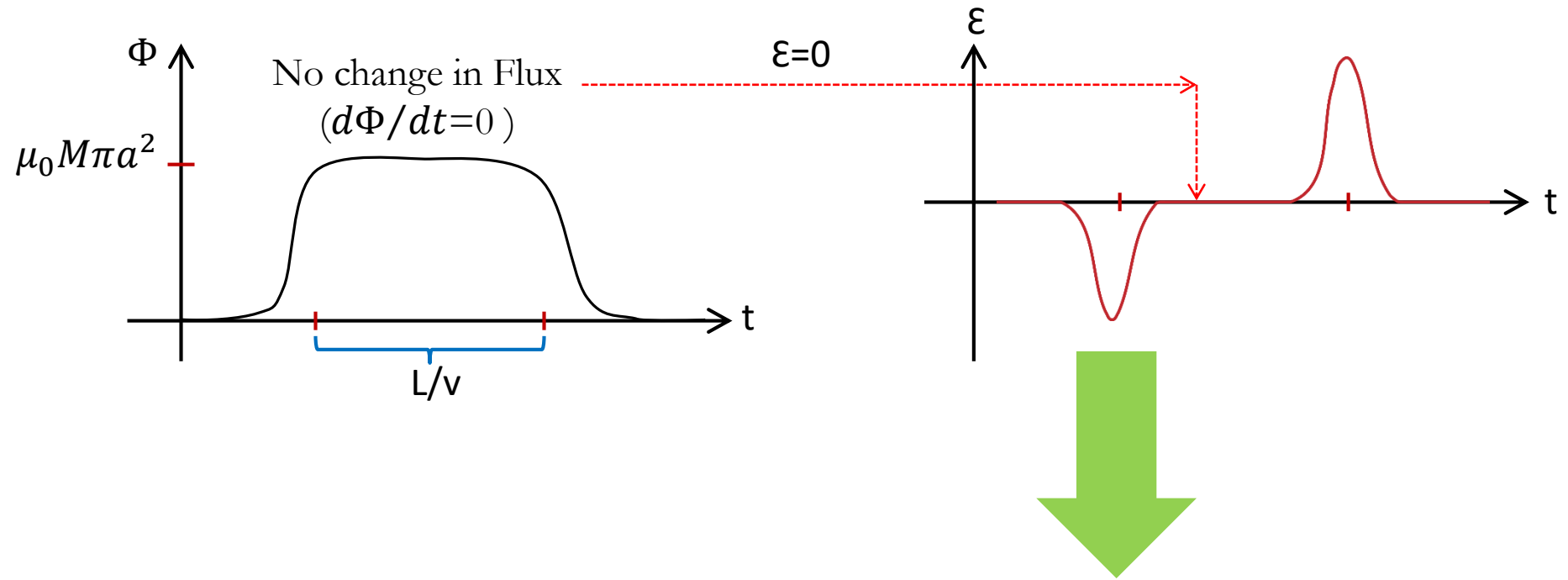


Figure 2b

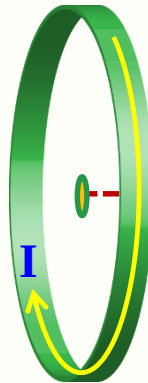
Assigning the Direction of current



Φ_{induced} will be -ve w.r.t Φ_{magnet}

Direction of Magnetic Field
generated by loop

Clockwise



Flux generated due to $\mathcal{E}$ is Φ_{induced} will be opposing the change in the Φ of magnet.

Lenz's law

Keeping track of the signs in Faraday's law can be a real headache.

For instance, in previous example we would like to know which way around the ring the induced current flows?

In principle, the right-hand rule does the job

We called Φ to be positive to the left, in Figure 1, so the positive direction for current in the ring is **counterclockwise**, as viewed from the left; since the first spike in Figure 2b is negative, the **first current pulse flows clockwise**, and **the second counterclockwise**.

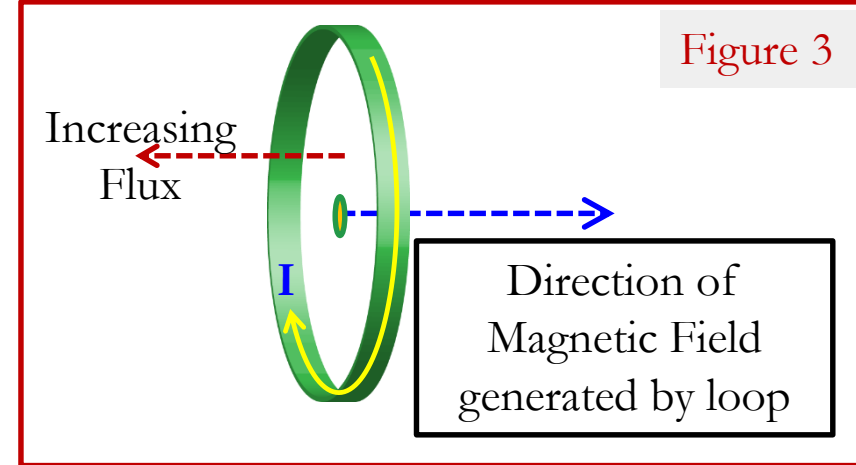
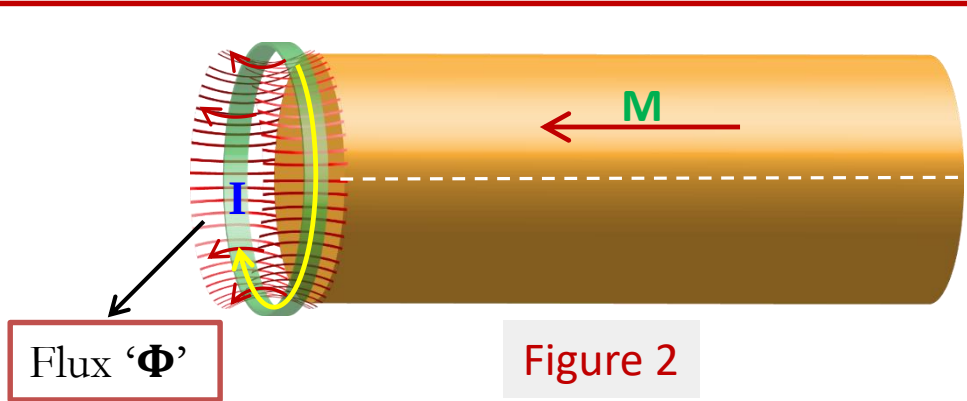
But there's a handy rule, called **Lenz's law**, whose sole purpose is to help you get the directions right.

The Lenz's law states; the induced current will flow in such a direction that the flux it produces tends to cancel the change.

Nature dislikes a change in flux

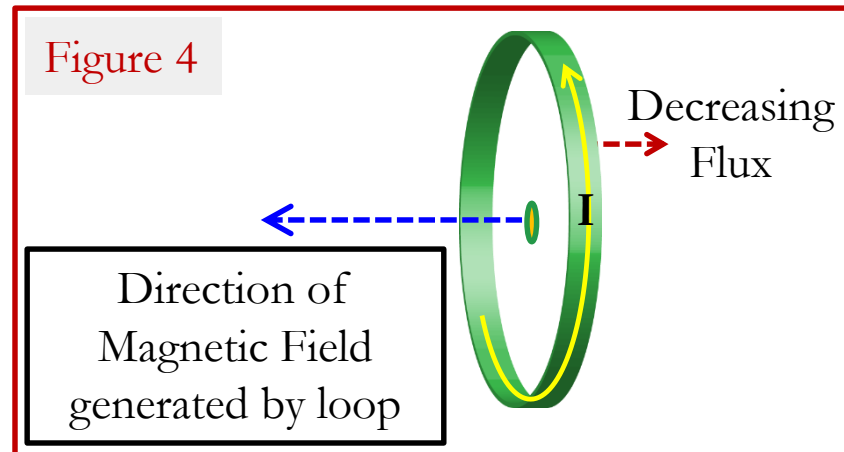
Using Lenz's law for Example 1

As the front end of the magnet in Example 1, enters the ring, the flux increases, so the current in the ring must generate a field to the right; current therefore in it will flow clockwise.



Notice that it is the change in flux, not the flux itself, that nature abhors.

When the tail end of the magnet exits the ring, the flux drops, so the induced current flows counterclockwise, in an effort to restore it.

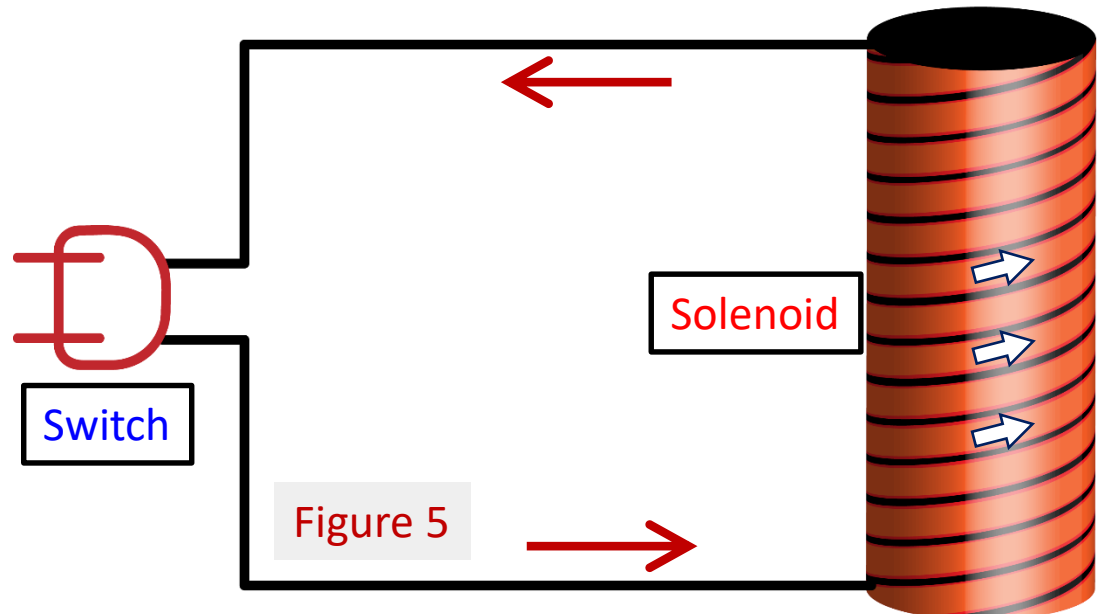
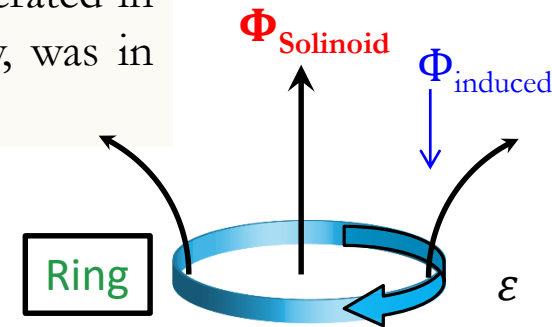


Example 2: The "jumping ring" demonstration. If you wind a solenoidal coil around an iron core (the iron is there to beef up the magnetic field), place a metal ring on top, and plug it in, the ring will jump several feet in the air (Figure 5). Why?

Solution: Initially when the switch is off, the flux through the ring was zero since the circuit does not contain any current.

Afterward a flux appeared (upward, in the diagram), and the emf generated in the ring led to a current (in the ring) which, according to Lenz's law, was in such a direction that its field tended to cancel this new flux.

This means that the current in the loop is opposite to the current in the solenoid. And opposite currents repel, so the ring flies off.



The Induced Electric Field

What Faraday's discovery tells us is that there are really **two** distinct kinds of **electric fields**: those attributable directly to **electric charges**, and those associated with **changing magnetic fields**.

The former can be calculated (in the static case) using Coulomb's law; the latter can be found by exploiting the analogy between Faraday's law and The Ampere's law,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

The Ampere's law,

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

Of course, the curl alone is not enough to determine a field--you must also specify the divergence. But as long as E is a pure Faraday field, due **exclusively to a changing B** (with **ρ free-charge density** = 0), Gauss's law says

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} = 0$$

While for magnetic fields, of course,

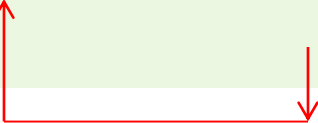
$$\nabla \cdot \vec{B} = 0$$

So the parallel is complete, and We conclude that Faraday's law of-induced electric fields are determined by $-(d\mathbf{B}/dt)$ in exactly the same way as magneto-static fields are determined by $\mu_0\mathbf{J}$.

In particular, if symmetry permits, we can use all the tricks associated with Ampere's law in integral form,

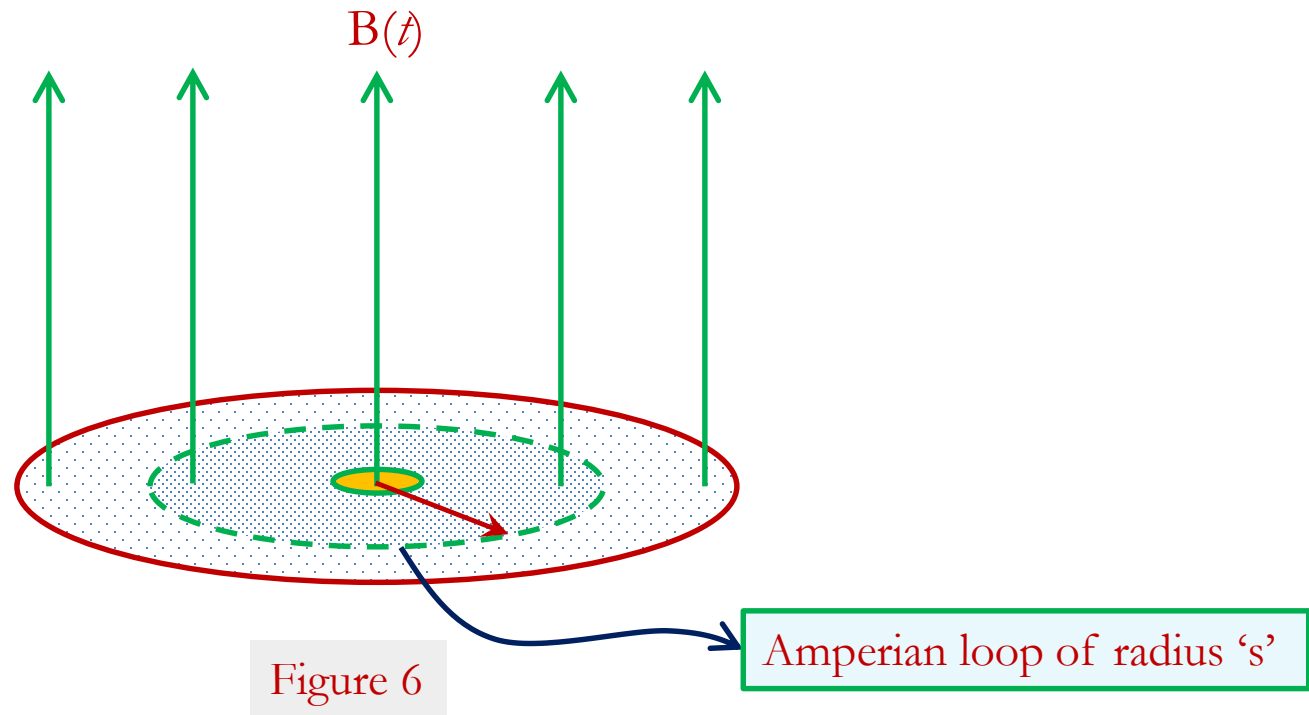
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enclosed}$$

But this time it's Faraday's law in integral form:

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d\Phi_{Mag}}{dt}$$


The rate of change of (magnetic) flux through the Amperian loop plays the role formerly assigned to $\mu_0 I_{enclosed}$.

Example 3: A uniform magnetic field $\mathbf{B}(t)$, pointing straight up, fills the shaded circular region of Figure 6. If $\mathbf{B}$ is changing with time, what is the induced electric field $\mathbf{E}_{\text{ind}}$?



Solution:

E will point in the circumferential direction, just like the magnetic field inside a long straight wire carrying a uniform current density.

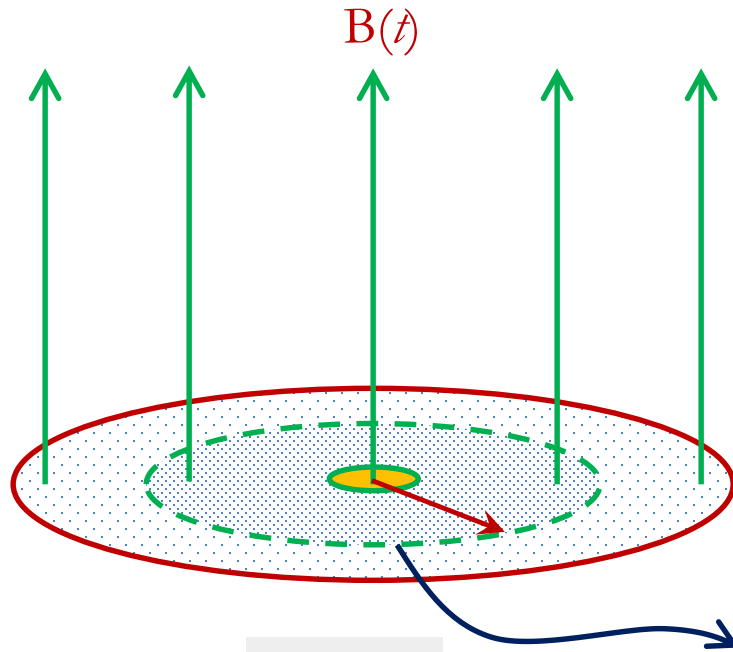


Figure 6

An Amperian loop of radius s is drawn as shown in figure 6, and applying the Faraday's law:

Amperian loop of radius ' s '

Faraday's law

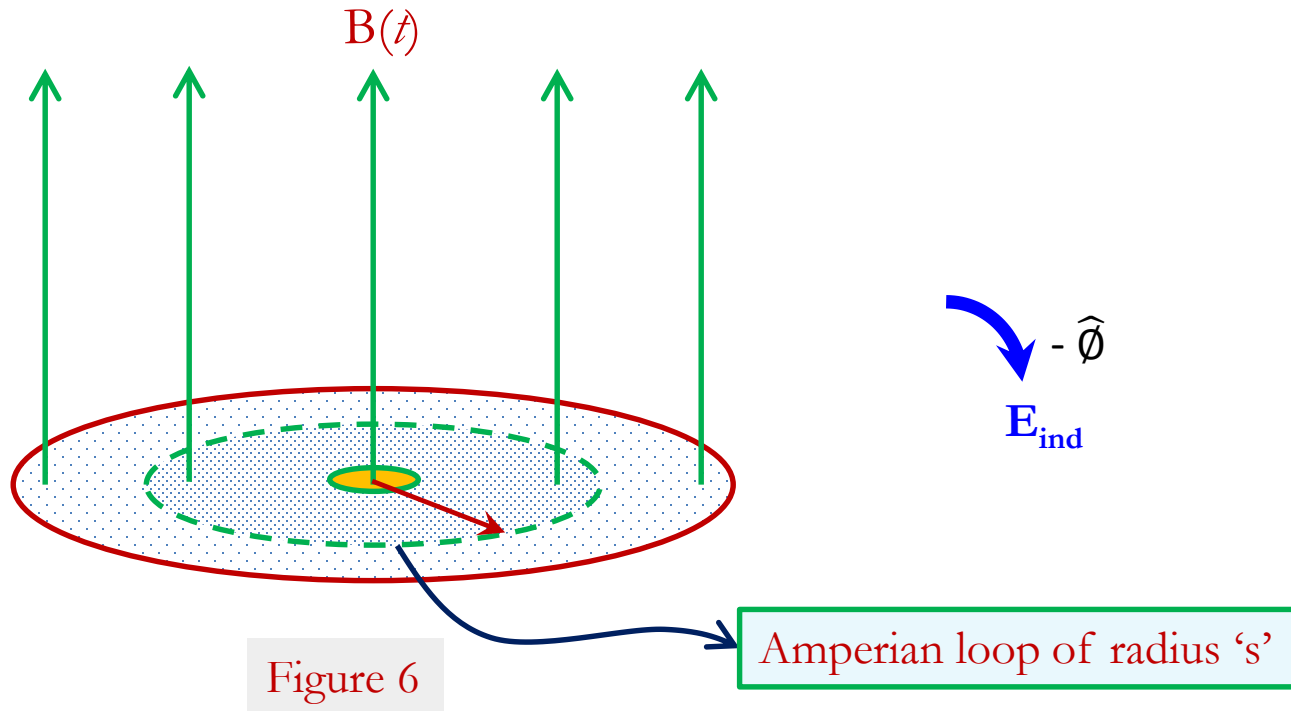
$$\oint \vec{E} \cdot d\vec{l} = E (2\pi s) = -\frac{d\Phi}{dt} = -\frac{d}{dt} (B(t) \pi s^2)$$

$$\oint \vec{E} \cdot d\vec{l} = -\pi s^2 \frac{dB}{dt}$$

Therefore

$$E = -\frac{s}{2} \frac{dB}{dt} \hat{\phi}$$

If B is increasing, $\mathbf{E}$ runs clockwise, as viewed from above.



Example 4: A line charge ' λ ', is glued onto the rim of a wheel of radius ' b ', which is then suspended horizontally as shown in **Figure 7**, so that it is free to rotate. In the central region, out to radius a , there is a uniform magnetic field $\mathbf{B}_0$, pointing up. Now someone turns the field off. What happens?

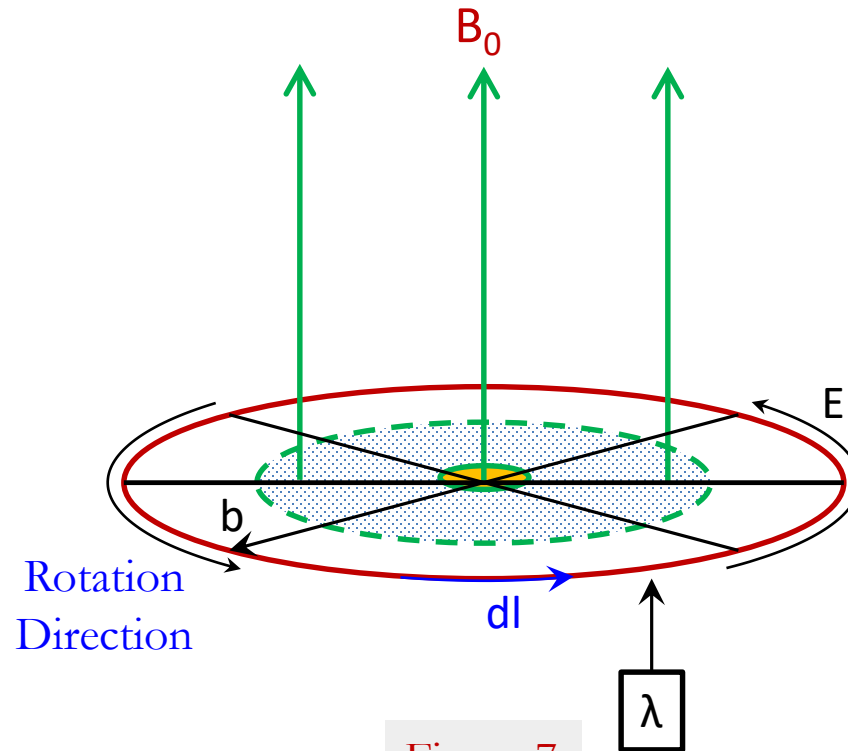


Figure 7

Solution:

The changing magnetic field will induce an electric field, **curling** around the axis of the wheel. This **electric field exerts a force on the charges at the rim**, and the **wheel starts to turn**. According to Lenz's law, it will rotate in such a direction that its field tends to restore the upward flux. The motion, then, is **counterclockwise**, as viewed from above.

Quantitatively, Faraday's law says

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}$$

B_0 decreasing
When switched off

Φ Induced
To restore this B_0
 Φ Induced
will act upwards

$$\Phi = \oint \vec{B} \cdot d\vec{a} = B\pi a^2$$

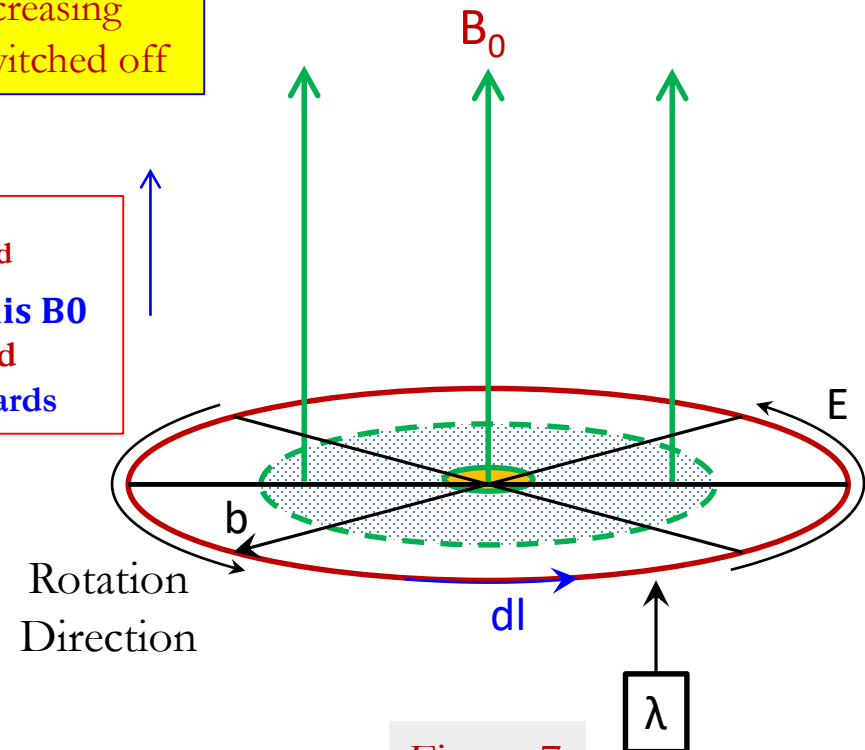


Figure 7

$$-\frac{d\Phi}{dt} = -\pi a^2 \frac{dB}{dt}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}$$

$$\Phi = \oint \vec{B} \cdot d\vec{a} = B\pi a^2$$

$$\oint \vec{E} \cdot d\vec{l} = -\pi a^2 \frac{dB}{dt}$$

Now, the torque on a segment of length $d\mathbf{l}$ is

$$d\vec{\tau} = \vec{r} \times \vec{F}$$

The force on the small element $d\mathbf{l}$ due to $\mathbf{E}$

$$\vec{F} = q\vec{E}$$

$$q = \lambda dl$$

Charge on the small element $d\mathbf{l}$

$$F = \lambda dl E$$

$$d\vec{\tau} = b\lambda dl E$$

Hence, the torque on the element is therefore

$$\vec{d\tau} = b\lambda dl E$$

Now, the total torque on the wheel is therefore

$$\oint \vec{d\tau} = \tau = \oint b\lambda dl E$$

$$\tau = b\lambda \oint E dl$$



$$\tau = -\pi a^2 b\lambda \frac{dB}{dt}$$

$$\oint \vec{E} \cdot \vec{dl} = -\pi a^2 \frac{dB}{dt}$$

Here the direction of E and dl are same
hence

$$\vec{E} \cdot \vec{dl} = E dl$$

The angular momentum imparted to the wheel

$$\text{Angular momentum}(L) = \int \tau dt = \int -\pi a^2 b \lambda \frac{dB}{dt} dt$$

$$= -\pi a^2 b \lambda \int_{B_0}^0 dB \quad \leftarrow \text{Final Magnetic field}$$

$$\tau = -\pi a^2 b \lambda \frac{dB}{dt}$$

$$= \pi a^2 b \lambda B_0$$

It doesn't matter how fast or slow you turn off the field; the ultimate angular velocity of the wheel is the same regardless.

It's the electric field that did the rotating

Magneto-statics and Faraday's Laws

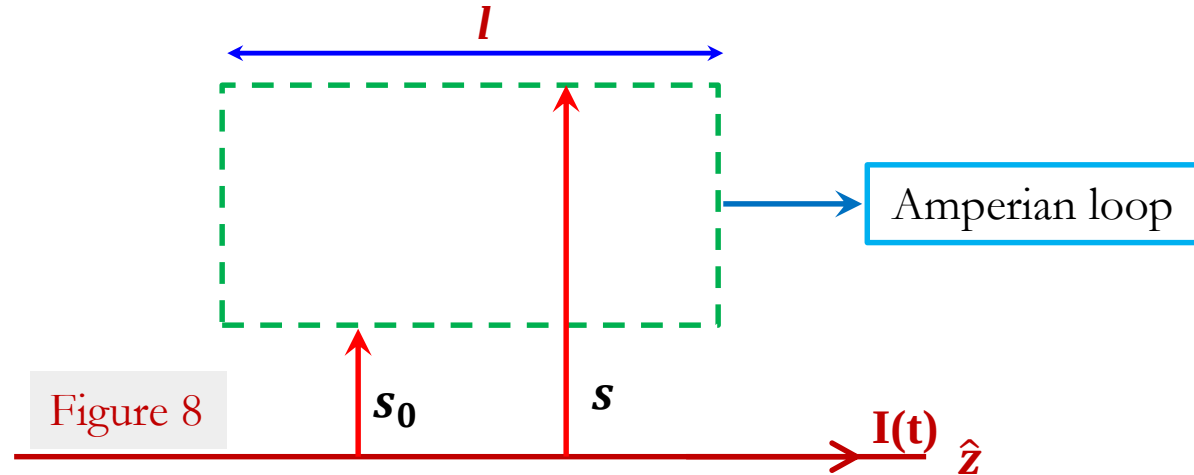
Electromagnetic induction occurs only when the magnetic fields are changing, and yet we would like to use the apparatus of magneto-statics i.e. Ampere's law, the Biot-Savart law, and the rest, to calculate those magnetic fields.

Technically, any result derived in this way is only approximately correct.

But in practice the error is usually **negligible** unless the field fluctuates extremely rapidly, or you are interested in points very far from the source.

This regime, in which magneto-static rules can be used to calculate the magnetic field on the right hand side of Faraday's law, is called **Quasistatic**.

Example 5: An infinitely long straight wire carries a slowly varying current $I(t)$. Determine the induced electric field, as a function of the distance 's' from the wire.



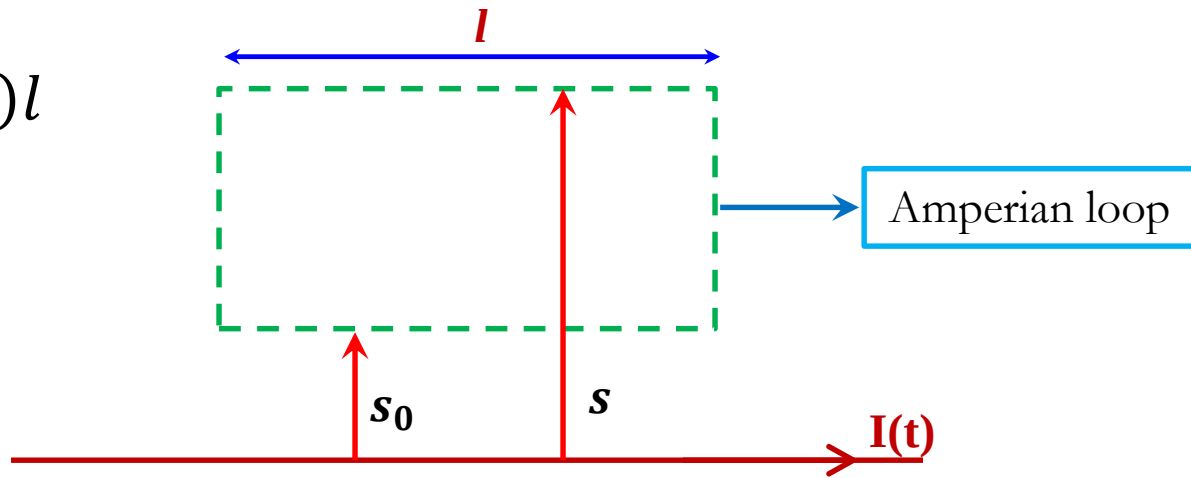
Solution: In the quasi-static approximation, the magnetic field is $(\mu_0 I / 2\pi s)$, and it circles around the wire. Like the **B-field** of a solenoid, **E** here runs parallel to the axis.

For the rectangular "Amperian loop" in Figure 8, Faraday's law gives:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}$$

$$\oint \vec{E} \cdot d\vec{l} = E(s_0)l - E(s)l$$

$$\Phi = \oint \vec{B} \cdot d\vec{a}$$



$$\oint \vec{E} \cdot d\vec{l} = E(s_0)l - E(s)l = -\frac{d}{dt} \oint \vec{B} \cdot d\vec{a}$$

Quasistatic approximation

$$B = \frac{\mu_0 I}{2\pi s}$$

Field due to long straight wire

$$\oint \vec{E} \cdot d\vec{l} = E(s_0)l - E(s)l = -\frac{d}{dt} \oint \frac{\mu_0 I}{2\pi s} \cdot d\vec{a}$$

$$\oint \vec{E} \cdot d\vec{l} = E(s_0)l - E(s)l = -\frac{\mu_0 l}{2\pi} \frac{dI}{dt} \int_{s_0}^s \frac{1}{s'} ds'$$

$$E(s_0)l - E(s)l = -\frac{\mu_0 l}{2\pi} \frac{dI}{dt} \int_{s_0}^s \frac{1}{s'} ds'$$

$$E(s_0)l - E(s)l = -\frac{\mu_0 l}{2\pi} \frac{dI}{dt} [\ln s - \ln s_0]$$

$$\mathbf{E}(\mathbf{s}) = \left[\frac{\mu_0}{2\pi} \frac{dI}{dt} \ln s + K \right] \hat{\mathbf{z}}$$

Equation 1

where K is constant (that is to say, it is independent of s but it might still be a function of t) for very far distances $S \rightarrow \infty$, from Eq - 1 $E \rightarrow \infty$ but in reality $E \rightarrow 0$

In this problem we have overstepped the limits of the **Quasistatic** approximation.

The **Quasistatic** approximation should hold only for $s \ll ct$; c = speed of light

Inductance

Suppose you have two loops of wire, at rest (Figure 9). If you run a steady current I_1 around loop 1, it produces a magnetic field B_1 . Some of the field lines pass through loop 2; let Φ_2 be the flux of B_1 through loop 2. It could be really difficult to calculate B_1 .

But a glance at the Biot-Savart law

$$\vec{B}_1 = \frac{\mu_0 I_1}{4\pi} \oint \frac{d\vec{l}_1 \times \hat{r}}{r^2}$$

It reveals one significant fact about this field: It is proportional to the current I_1 .

Therefore, the flux through loop 2

$$\Phi_2 = \oint \vec{B}_1 \cdot d\vec{a}_2$$

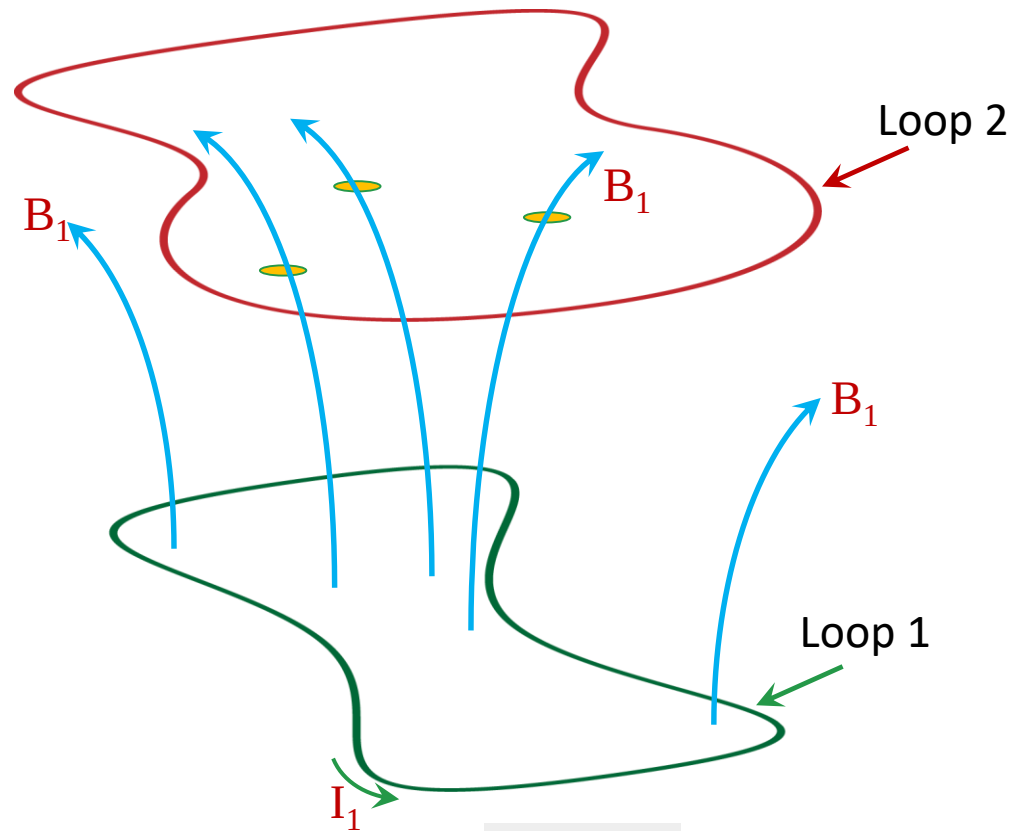


Figure 9

Hence

$$\Phi_2 = M_{21} I_1$$

Here, M_{21} is the constant of proportionality; it is known as the mutual inductance of the two loops.

There is a cute formula for the mutual inductance, which you can derive by expressing the flux in terms of the vector potential and invoking Stokes' theorem:

$$\Phi_2 = \int \vec{B}_1 \cdot d\vec{a}_2$$

$$\Phi_2 = \int (\nabla \times \vec{A}_1) \cdot d\vec{a}_2$$

$$\Phi_2 = \oint \vec{A}_1 \cdot d\vec{l}_2$$

We know from vector potential

$$\vec{A}_1 = \frac{\mu_0 I_1}{4\pi} \oint \frac{dl_1}{r}$$

Hence

$$\Phi_2 = \frac{\mu_0 I_1}{4\pi} \oint \oint \frac{dl_1}{r} \cdot \vec{dl}_2$$

$$\Phi_2 = M_{21} I_1$$

Evidently

$$M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{dl_1 \cdot \vec{dl}_2}{r}$$

Neumann formula

Some important points about M_{21} (Neumann formula)

- ❑ M_{21} is a purely geometrical quantity, having to do with the sizes, shapes, and relative positions of the two loops.
- ❑ The integral in Neumann's formula is unchanged if we switch the roles of loops 1 and 2; it follows that

$$M_{21} = M_{12}$$

This is an astonishing conclusion: Whatever the shapes and positions of the loops, the flux through loop-2 when we run a current 'I' around loop-1 is identical to the flux through 'I' when we send the same current 'I' around loop-2. We may as well drop the subscripts and call them both **M**.

Self Inductance

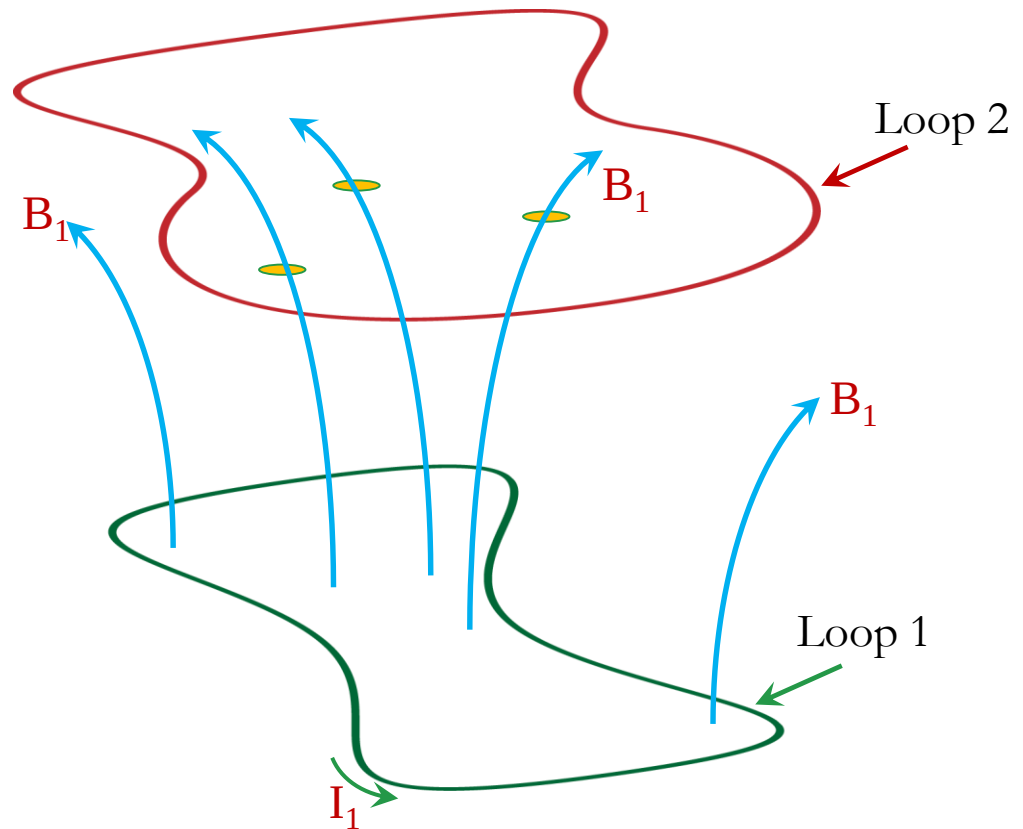
Suppose now that you vary the current in **loop 1**. The flux through loop 2 will vary accordingly, and Faraday's law says this changing flux will **induce an emf** in **loop 2**:

$$\varepsilon_2 = -\frac{d\Phi_2}{dt}$$

$$\varepsilon_2 = -M \frac{dI_1}{dt}$$

since

$$\Phi_2 = MI_1$$



What a remarkable thing: Every time you change the current in **loop-1**, an induced current flows in **loop-2**, even though **there are no wires connecting them!**

A changing current not only induces an emf in any nearby loops, it also induces an emf in the source loop itself (**Figure 11**). Once again, the field (and therefore also the flux) is proportional to the current:

$$\Phi_1 = LI$$

The constant of proportionality 'L' is called the **self-inductance** (or simply the **inductance**) of the loop.

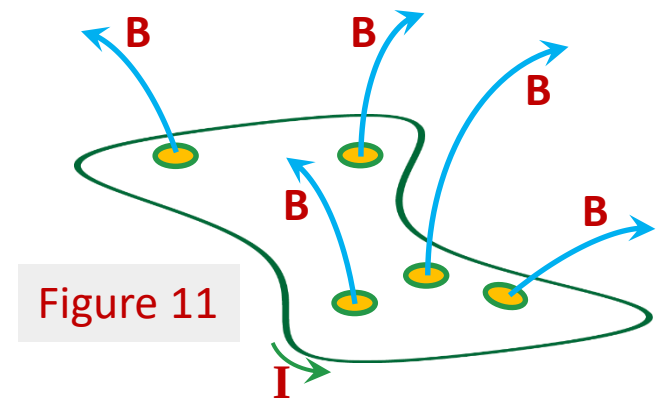


Figure 11

As with M , it depends on the geometry (**size and shape**) of the loop. If the current changes, the emf induced in the loop is:

$$\varepsilon = -L \frac{dI}{dt}$$

$$\Phi_1 = LI$$

Inductance is measured in **Henries (H)**; **1 H = volt-second per ampere**.

Inductance (like capacitance) is an intrinsically positive quantity. Lenz's law, which is enforced by the minus sign in the above Equation, dictates that the emf is in such a direction as to oppose any change in current. For this reason, it is called a **back emf**. Whenever you try to alter the current in a wire, you must fight against this back emf.

Thus inductance plays somewhat the same role in electric circuits that mass plays in mechanical systems: The **greater 'L'** is, the **harder it is to change the current**, just as the larger the mass, the harder it is to change an object's velocity.

Energy in Magnetic Fields

The work we must have to do against the **back emf** to get the current going.

This is a fixed amount, and it is recoverable: you get it back when the current is turned off. In the meantime it represents energy latent in the circuit; as we'll see in a moment, it can be regarded as energy stored in the magnetic field.

The work done on a unit charge, against the back emf, in one trip around the circuit is $-\varepsilon$ (the minus sign records the fact that this is the work done by you against the emf, not the work done by the emf). The amount of charge per unit time passing down the wire is 'I'. So the total work done per unit time is:

$$\frac{dW}{dt} = -\varepsilon I = L \frac{dI}{dt} I$$

Integrating over time

$$W = \frac{1}{2} L I^2$$

Here we start with zero current and build it up to a final value 'I'.

It depends only on the geometry of the loop and current

$$\Phi = \int_S \vec{B} \cdot \vec{da}$$

$$\Phi = \int_S \nabla \times \vec{A} \cdot \vec{da}$$

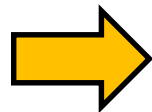
$$\Phi = \oint_c \vec{A} \cdot \vec{dl}$$

$$\Phi = LI$$

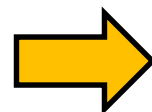
$$LI = \oint_c \vec{A} \cdot \vec{dl}$$

Hence

$$W = \frac{1}{2} LI^2$$



$$W = \frac{1}{2} I \oint \vec{A} \cdot \vec{dl}$$



$$W = \frac{1}{2} \oint \vec{A} \cdot \vec{I} dl$$

The generalization to volume currents is obvious:

$$W = \frac{1}{2} \oint_V \vec{A} \cdot \vec{J} d\tau$$

But we can do even better, and express W entirely in terms of the magnetic field:

The Ampere's law,

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

Let's eliminate $\vec{J}$:

$$W = \frac{1}{2\mu_0} \oint_V \vec{A} \cdot \nabla \times \vec{B} d\tau$$

Equation 1

The vector product rule:

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$$



$$\vec{A} \cdot (\nabla \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \nabla \cdot (\vec{A} \times \vec{B})$$

So

$$W = \frac{1}{2\mu_0} \oint_V \vec{A} \cdot \nabla \times \vec{B} \, d\tau$$

$$\vec{A} \cdot (\nabla \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \nabla \cdot (\vec{A} \times \vec{B})$$

Since

$$\nabla \times \vec{A} = \vec{B}$$

$$\vec{A} \cdot (\nabla \times \vec{B}) = \vec{B} \cdot (\vec{B}) - \nabla \cdot (\vec{A} \times \vec{B})$$

Substituting this value back to the equation 1

$$W = \frac{1}{2\mu_0} \oint_V \vec{B} \cdot (\vec{B}) - \nabla \cdot (\vec{A} \times \vec{B}) \, d\tau$$

$$W = \frac{1}{2\mu_0} \oint B^2 d\tau - \oint \nabla \cdot (\vec{A} \times \vec{B}) dl$$

$$W = \frac{1}{2\mu_0} \oint B^2 d\tau - \oint_S (\vec{A} \times \vec{B}) dl$$

if we agree to integrate over all space, then the surface integral goes to zero, and we are left with (total volume (∞) and no surface)

$$W = \frac{1}{2\mu_0} \int_{all\ space} B^2 d\tau$$

In view of this result, we say the energy is "stored in the magnetic field," in the amount $(B^2/2\mu_0)$ per unit volume.

In the light of this, it is extraordinary how similar the magnetic energy formulas are to their electrostatic counterparts:

$W = \frac{1}{2} \int (V\rho) d\tau$	$W = \frac{\epsilon_0}{2} \int E^2 d\tau$
Electrostatics	
Magneto-statics	
$W = \frac{1}{2} \int (\vec{A} \cdot \vec{j}) d\tau$	$W = \frac{1}{2\mu_0} \int B^2 d\tau$